

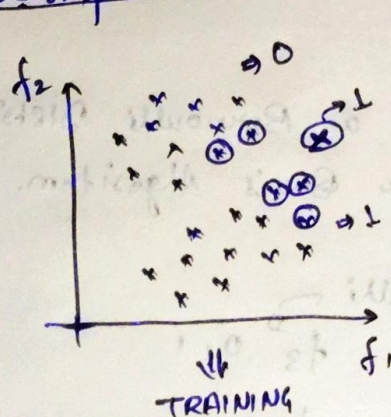
* K-Nearest Neighbour (KNN)

M-80

- I. Classification
- II. Regression

I. Classification:

[Binary category]



f_1	f_2	y
-	-	0
-	-	1
0.8		2
1.3		3

I. We have to initialize the K value

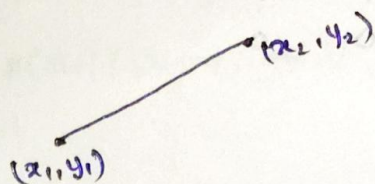
$$K > 0 \dots \infty$$

$K = 1, 2, 3, 4, 5 \dots$ Hyperparameter

II. Find The K Nearest Neighbour for the Test Data

III. From those $K > 5$ how many neighbours belong to 0 category and 1 category.

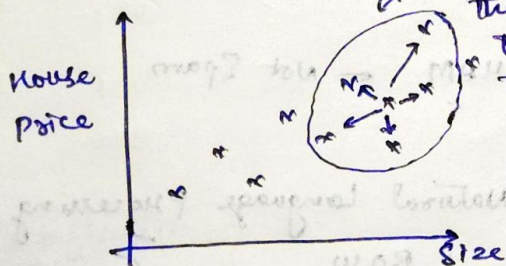
* Euclidian Distance



$$\text{Distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Median

II. Regression:

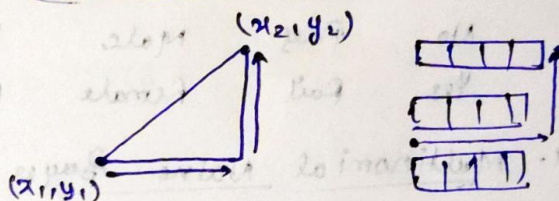


Time complexity $O(n)$ of million of datapoints

{KDTree {optimize
{Ball Tree}}

Binary Tree

* Manhattan Distance



Note:

All the steps are same in both Classification & Regression find K value based on the neighbours.

Find out the average or you have lots of outliers you can go also with the median.

Note: In Binary Tree it is calculate the distance between every point in the test data is reduced by half because it will not be checking with respect to every data point. That is the power of Binary Tree.